H2 Math Summary

Example Questions

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1 Graphing Techniques

1 The function f is given by

$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

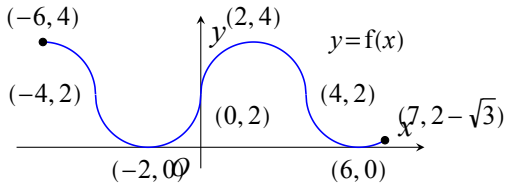
(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

The curve C has equation $\dfrac{(y-3)^2}{a^2}-(x+2)^2=1$, where a is a positive real constant.

(c) State the equations of the asymptotes of C in terms of a. [1]

(d) Determine the range of values of a if there is at most one intersection between C and the graph of $y=\mathrm{f}(x)$. [2]

[VJC 2025 P1 Q5]

Solution		Comments
(a) **[3]**		
(b) **[1]**	$\text{Area} = 2\times 4 + \frac{1}{2}\pi(2^2)$ $= 8 + 2\pi$	
(c) **[1]**	$\frac{(y-3)^2}{a^2} = (x+2)^2$ $y - 3 = \pm a(x+2)$ $y = 3 + a(x+2)$ and $y = 3 - a(x+2)$	
(d) **[2]**	C is a hyperbola with centre $(-2, 3)$ and vertices $(-2, 3+a)$, $(-2, 3-a)$ When $a = 3$, exactly one intersection at $(-2, 0)$ $\therefore a \geqslant 3$	

2 **(a)** A curve C with equation $y = \mathrm{f}(x)$ undergoes in succession, the following transformations.

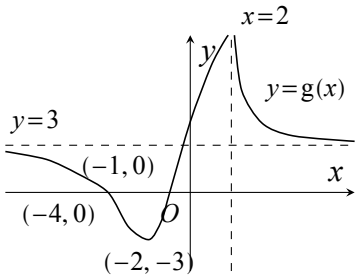
A: A reflection in the x-axis.

B: A stretch parallel to the x-axis with scale factor $\frac{1}{2}$, with the y-axis invariant.

The resulting curve has equation $y = ax^2 + \dfrac{b}{x}$, where a and b are real constants.

Given that $\left(-1, \frac{1}{3}\right)$ is a turning point of $y = \dfrac{1}{\mathrm{f}(x)}$, find the values of a and b and state the equation of C. [5]

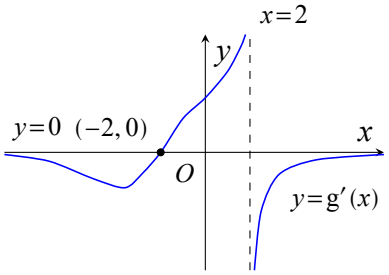
(b) The diagram below shows the curve of $y = \mathrm{g}(x)$. The curve has a minimum point at $(-2, -3)$ and crosses the x-axis at $(-1, 0)$ and $(-4, 0)$. The line $x = 2$ is the vertical asymptote and the line $y = 3$ is the horizontal asymptote.



(i) Sketch the graph of $y = \mathrm{g}'(x)$, labelling the coordinates of all relevant point(s) and state the equations of any asymptotes. [2]

(ii) Find the area of the region bounded by the graph of $y = \mathrm{g}'(x)$, the lines $x = -4$, $x = -2$ and the x-axis. [2]

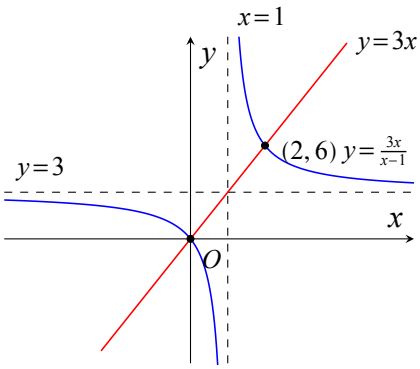
[VJC 2025 P1 Q8]

Solution		Comments
(a) **[5]**	$y=ax^2+\dfrac{b}{x}$	
	$\xrightarrow{B^{-1}:\ x\to\frac{x}{2}}\quad y=a\left(\dfrac{x}{2}\right)^2+\dfrac{b}{\frac{x}{2}}=\dfrac{ax^2}{4}+\dfrac{2b}{x}$	
	$\xrightarrow{A^{-1}:\ y\to -y}\quad y=-\dfrac{ax^2}{4}-\dfrac{2b}{x}=\mathrm{f}(x)$	
	$\left(-1,\frac{1}{3}\right)$ on $y=\frac{1}{\mathrm{f}(x)}\ \Rightarrow\ (-1,3)$ is a turning point on $y=\mathrm{f}(x)$	
	$-\dfrac{a}{4}+2b=3$	$---(1)$
	$\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{1}{2}ax+\dfrac{2b}{x^2}$	
	$x=-1,\ \dfrac{\mathrm{d}y}{\mathrm{d}x}=0:\quad \dfrac{1}{2}a+2b=0$	$---(2)$
	Using GC, $a=-4,\ b=1$	
	$\therefore C:\ y=x^2-\dfrac{2}{x}$	
(b)(i) **[2]**		
	Asymptotes: $x=2,\ y=0$	
(b)(ii) **[2]**	Required area $=-\int_{-4}^{-2}\mathrm{g}'(x)\,\mathrm{d}x$	
	$=-\left[\mathrm{g}(x)\right]_{-4}^{-2}$	
	$=-\left[\mathrm{g}(-2)-\mathrm{g}(-4)\right]$	
	$=-\left[-3-0\right]$	
	$=3$ units^2	

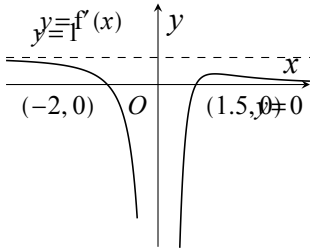
3 (a) (i) On a single diagram, sketch the graphs of $y=\dfrac{3x}{x-1}$ and $y=3x$, stating the equation(s) of any asymptote(s) and the coordinates of points of intersection. [4]

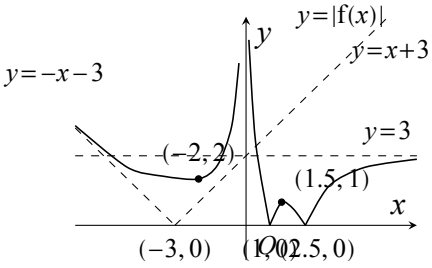
(ii) Hence find the area bounded by the curve $y=\dfrac{3x}{x-1}$ and the lines $y=3x$ and $y=8$. [2]

[SAJC 2025 P1 Q7]

Solution		Comments
(a)(i) **[4]**	 $\dfrac{3x}{x-1}=3x$ $3x=3x(x-1)$ $3x(x-2)=0$ $x=0$ or $x=2$ $\therefore$ intersections: $(0,0)$, $(2,6)$ Asymptotes: $x=1$, $y=3$	
(a)(ii) **[2]**	$y=\dfrac{3x}{x-1} \Rightarrow x=\dfrac{y}{y-3}$ $y=3x \Rightarrow x=\dfrac{y}{3}$ $\text{Area}=\int_6^8 \left(\dfrac{y}{3}-\dfrac{y}{y-3}\right)\,\text{d}y$ $=1.13 \text{ units}^2$ (3 s.f.)	

4 The graphs of $y=\mathrm{f}'(x)$ and $y=|\mathrm{f}(x)|$ are shown below.





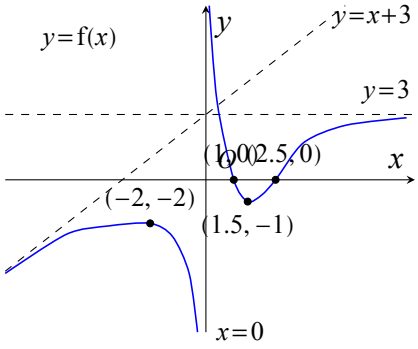
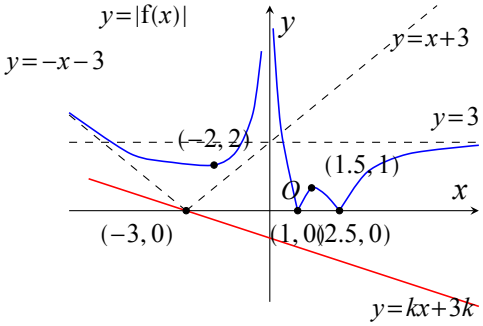
(a) State the nature of all turning point(s) of the graph of $y=\mathrm{f}(x)$. [2]

(b) State the range of values of x where f is decreasing. [2]

(c) Sketch the graph of $y=\mathrm{f}(x)$ indicating clearly the equations of the asymptotes, coordinates of the turning point(s) and the intersections with the axes. [3]

(d) On the copy of the graph of $y=|\mathrm{f}(x)|$ in the Printed Answer Booklet, sketch and label a line $y=kx+3k$, where k is a constant. Hence state the range of values of k for which there is no real solution to the equation $|\mathrm{f}(x)|=kx+3k$. [2]

[DHS 2025 P1 Q5]

Solution		Comments
(a) **[2]**	$x=-2$: $f'(x)$ changes from + to − $\Rightarrow$ maximum point $x=1.5$: $f'(x)$ changes from − to + $\Rightarrow$ minimum point	
(b) **[2]**	$f'(x) \leqslant 0$ $-2 \leqslant x < 0$ or $0 < x \leqslant 1.5$	
(c) **[3]**	 Asymptotes: $x=0$, $y=3$, $y=x+3$	
(d) **[2]**	 $y=kx+3k=k(x+3)$ passes through $(-3, 0)$ $\therefore\ -1 \leqslant k < 0$	

5 A curve C is defined parametrically by

$$x = a\tan t, \qquad y = a\sec^2 t \sin t, \qquad -\frac{\pi}{4} < t < \frac{\pi}{4},$$

where a is a positive constant.

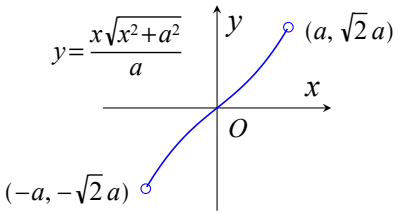
(a) Show that $\dfrac{\tan t}{\sqrt{1+\tan^2 t}} = \sin t$. [1]

(b) By using part **(a)** or otherwise, find the cartesian equation of C in the form $y = \mathrm{f}(x)$, simplifying your answer. [2]

(c) Show that $\mathrm{f}(-x) = -\mathrm{f}(x)$. Hence sketch C. [2]

(d) Find the exact area of the region bounded by C, the x-axis and the lines $x = -A$ and $x = A$, where $0 < A < a$, leaving your answer in terms of A and a. [3]

[DHS 2025 P1 Q6]

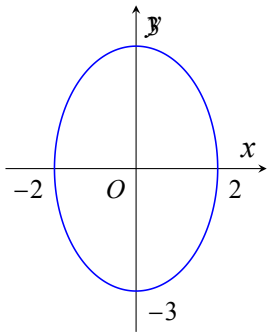
Solution		Comments
(a) **[1]**	$\text{LHS} = \dfrac{\tan t}{\sqrt{1+\tan^2 t}} = \left(\dfrac{\sin t}{\cos t}\right)\left(\dfrac{1}{\sec t}\right)$ $= \left(\dfrac{\sin t}{\cos t}\right)(\cos t)$ $= \sin t = \text{RHS (shown)}$	
(b) **[2]**	$x = a\tan t \Rightarrow \tan t = \dfrac{x}{a}$ $y = a\sec^2 t \sin t$ $= a\left(\tan^2 t+1\right)\left(\dfrac{\tan t}{\sqrt{1+\tan^2 t}}\right)$ $= a\left(\left(\dfrac{x}{a}\right)^2+1\right)\left(\dfrac{\frac{x}{a}}{\sqrt{1+\left(\frac{x}{a}\right)^2}}\right)$ $= x\sqrt{1+\left(\dfrac{x}{a}\right)^2}$ $= \dfrac{x\sqrt{x^2+a^2}}{a}, \quad \text{where } -a<x<a$	
(c) **[2]**	$\text{f}(-x) = \dfrac{-x\sqrt{(-x)^2+a^2}}{a}$ $= -\dfrac{x\sqrt{x^2+a^2}}{a}$ $= -\text{f}(x) \text{ (shown)}$  	
(d) **[3]**	$\text{Area} = 2\int_0^A \dfrac{x\sqrt{x^2+a^2}}{a}\,\text{d}x \quad \text{(by symmetry)}$ $= \dfrac{1}{a}\int_0^A 2x\sqrt{x^2+a^2}\,\text{d}x$ $= \left(\dfrac{1}{a}\right)\left[\dfrac{2}{3}\left(x^2+a^2\right)^{\frac{3}{2}}\right]_0^A$ $= \dfrac{2}{3a}\left(\left(A^2+a^2\right)^{\frac{3}{2}} - a^3\right)$	

6 A complex number z varies with t such that

$$z = 2\cos t + \mathrm{i}(3\sin t), \quad \text{where } 0 \leqslant t < 2\pi.$$

(a) By taking $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$, sketch on an Argand diagram the curve that shows the positions of the points representing the complex number z. Find the cartesian equation of this curve. [3]

[DHS 2025 P1 Q8]

Solution		**Comments**
(a) **[3]**	 $x = 2\cos t, \quad y = 3\sin t$ $\cos t = \frac{x}{2}, \quad \sin t = \frac{y}{3}$ $\cos^2 t + \sin^2 t = 1$ $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$	

7 A curve C has equation $y=x+\dfrac{9}{x+1}$.

(a) Using differentiation, find the range of values of x such that C concaves upwards. [2]

(b) Sketch C. [3]

[DHS 2025 P2 Q1]

Solution		Comments
(a) **[2]**	$y=x+\dfrac{9}{x+1}$ $\dfrac{dy}{dx}=1-\dfrac{9}{(x+1)^2}$ $\dfrac{d^2y}{dx^2}=\dfrac{18}{(x+1)^3}$ C concaves upwards $\Rightarrow \dfrac{18}{(x+1)^3}>0$ $(x+1)^3>0 \Rightarrow x>-1$	
(b) **[3]**	$\dfrac{dy}{dx}=0$: $(x+1)^2=9$ $x+1=\pm 3 \Rightarrow x=2$ or $x=-4$ Stationary points: $(-4,-7)$, $(2,5)$ $x=0$: $y=9$ Asymptotes: $x=-1$, $y=x$ 